Runway

The complete guide: what we built, how it works, and how to show it

Sheridan Datathon 2026 · Tangerine track · Team Runway (Sukhman Aulakh, engine lead)

Written November 8, 2026, from the code on main and evidence.json. Every number in this guide is a tested result from the data, not an estimate.

Live: https://runway-1085585470420.us-central1.run.app (delete after judging: docs/DEPLOY.md, section 4).

How to read this guide

Part	For	What you get
1. The simple version	Anyone, including you at 2 a.m.	What Runway does, explained with one customer as the example
2. The words	Anyone	Every term on screen in plain English
3. The product, screen by screen	You, before the demo	What each screen shows, what to click, what to say
4. The business logic	Bankers, consultants	Who gets which action, and why it makes money or saves it
5. The technical guide	Engineers, the professor	Architecture, data, every calculation, statistics, AI, API, tests, Docker and cloud
6. What you built	You, for Q&A and interviews	The work, round by round
7. Showing it to the judges	You, on the day	The 5-step pitch, the live twist, what to touch, the hard questions
8. Is it ready to ship?	You and a CTO	Checklist, what is and isn't production-grade
9. Strict rating and ranking	You	Every component scored, and where we honestly stand

1. The simple version

The idea in one sentence

> Banks look at your **average** balance. Overdrafts happen at your **lowest** balance. Runway watches the low point, finds the customers who are about to go below $0 before they do, and tells the bank exactly what to do about each one.

A story with one real customer: account 93172

Set the Time Machine to December 2024 and open customer 93172. Here is what the bank sees, step by step.

1. **Looks fine.** Their average balance over 6 months is **$18,503**. The standard runway formula (the one the judges gave us) says **no shortfall (120+ months)**. On paper, this person is safe forever.
2. **The low point.** But inside the month their balance dipped to **$238**. Lowest ÷ average is about **1%**: riskier than **98%** of customers. A banker would call it a **99% haircut**: almost all of the average is gone at the bottom of the month.
3. **What changed.** That same month the bank lent them **$166,958**, with a payment of **$4,638 a month**. The standard formula can't see it yet, because it only looks at past spending. Our realistic runway counts it: **2.4 months**, not 120+.
4. **What happened next.** They went overdrawn in **January 2025**. Our calibrated forecast had given them a **13.8%** chance of a first overdraft in the next 12 months (about 8 times the 1.6% average).
5. **What Runway tells the advisor to do.** Money in fell 27% ($14,323 a month vs $19,627 before), so: **call first this week**. Then a free alert when the balance falls below **$6,527** (their own typical dip, not a generic $100), and an automatic transfer from their own savings of the same size. A small overdraft limit comes last, and only after a call, because it costs them money (about **$108** for a month at Tangerine's own terms).
6. **And the loan?** A simple check at approval, using only data up to November 2024, would have **held that loan for review**: they were already in the riskiest 10%.

Why that matters, in three numbers

- If the bank flags the riskiest **10%** by the low point, it catches **34 of 55** first overdrafts the next year (62%). This held in 2023, 2024 and 2025, each tested on a year the model never saw.
- Every one of the **76** borrowers who was ever overdrawn defaulted on their loan. Overdrafts are an early warning for credit losses.
- A review rule at loan approval holds **17%** of loans; those loans default **34.7%** of the time vs **6.2%** for the rest.

What Runway is not

- It is not an automatic credit decision. A human advisor acts on every output.
- It does not use AI to calculate. Every number comes from the engine; AI only reads twist cards and writes sentences, and every number it writes is checked.
- It is built on 1990s Czech bank data (re-dated to 2020–2025), not real Tangerine customers. That's why a real rollout starts in shadow mode.

2. The words you need

Word on screen	Plain English	Example (93172, Dec 2024)
Money in / money out	Everything that arrived in the account in a month / everything that left it	About $18,301 in and a similar amount out per month
Standard runway	The judges' formula: months until savings run out after a shock, using medians. We never change it (it's the answer key)	No shortfall (120+)
Realistic runway	Our version: last 12 months, EI capped at $729/week and ending, pensions kept, the in-month dip, and new loan payments counted	2.4 months
Liquidity trough	Lowest balance ÷ average balance over 6 months. Near 0% = the account keeps dipping close to $0	1%
Haircut	1 − trough: the share of the average balance that's gone at the low point (bankers' word)	99%
Riskiest 10%	Customers whose trough is in the bottom 10% of all customers with 12+ months of history	Riskier than 98%
Calibrated risk	The chance of a first overdraft in 12 months, from the simulation, corrected so 1.6% really means 1.6%	13.8%
Twist card	The judges' surprise scenario (job loss, rent up, a repair). We turn its words into settings	Job loss + bills +15%
Verdict	Yes / No / Only on paper / Fragile: can they make it 6 months, and the biggest reason	—
Simulated futures	2,000 possible next years, built from the customer's own past months	“7 in 10 futures stay above $0”
Cushion / buffer	Money that should sit ready to cover the in-month dip (an auto-transfer or a small limit)	$6,527 (conservative $10,647)
Alert level	When the balance falls below this, warn the customer. Set at their own dip	$6,527
Tier	Which group a customer is in: already overdrawn, act now, watch, stable, new	Act now
Hazard ratio	How many times faster something happens to one group than another	Flagged: 11–12× faster first overdrafts
AUC	How well a score ranks risky above safe customers. 0.5 = coin flip, 1.0 = perfect	0.86 for the trough in 2025
Out of time	Built on one year, tested on the next. The honest way to test a forecast	Fit on 2024, judged on 2025

3. The product, screen by screen

Runway is one web app with five screens down the left side. Across the top: the **account box** (type any of 4,500 accounts), the **Time Machine** (Today = Dec 2025, Dec 2024, Dec 2023: every number is cut to that date, so nothing looks into the future), **Pitch mode**, and the **AI badge** (Gemini on Vertex AI, or offline rules).

01 Who's at risk (the bank's view)

What you see	What it means	Engine source
Headline numbers	Riskiest 10% → 34 of 55 overdrafts; 13–18× likelier; alert 3 months early	evidence.json → trough, impact
The whole book	Every customer in one tier with one action and the rate that tier really overdrew the next year	book_tiers, evidence → book
Advisor call list	The riskiest customers this month, each with their own cause and next step; click a row to open them	watchlist + diagnose
If everyone lost their job	Share of all customers still above $0 by month: standard vs realistic	portfolio_curve
Check the trough before lending	The loan-approval rule and its backtest	evidence → loan_policy
Who the riskiest are	Not poor: higher money in, more loans, almost no pensioners	evidence → segment

02 Know the customer

What you see	What it means	Engine source
4-step story strip	Looks fine → the low point → what changed → act. Builds itself for any account	build_context
Chance of a first overdraft	Calibrated 12-month risk (only for customers never overdrawn, 12+ months)	overdraft_risk
Liquidity trough card	Trough, percentile, haircut	liquidity_trough
Standard vs realistic runway	The judges' number next to ours	runway, realistic_runway
Why they're at risk, and what to do	Causes in order of urgency, each with an action sized from their own numbers	diagnose
When?	Among customers like them, how many had a first overdraft within 6 / 12 / 24 months	when_overdraft (survival)
Advisor assistant	Brief me before I call; draft a message to the customer in English or French	ai.py (words only)
Balance chart and what happened next	The real months; in Time Machine mode, what actually happened after	history, what_happened_next
Loan card	Their loan, its payment, and whether the approval check would have held it	loan_check

03 Stress test (the live twist)

1. Type or pick a twist card (for example “They lose their job and rent goes up 15%”).
2. Check the **chips**: the settings the AI read from the card. If the AI and the offline rules disagree on a field, it's flagged. Confirm before running.

3. Read the **verdict**: can they make it 6 months? Yes / No / Only on paper / Fragile, and **the biggest reason in months**.
4. Below: both runways, why they differ, the simulated futures (fan chart), and for risk teams an exposure profile (labelled uncalibrated).

04 Make a plan

- **Do this first**: the cause (for 93172, money in fell: call this week).
- **Then cover the dip**: 1. alert (free), 2. auto-transfer from savings (free), with a conservative option, 3. move money to chequing now. The cost of a limit at Tangerine's terms is shown, which is why it comes last.
- **Does a dip-sized cushion work?** The 2025 backtest, plus the risk-appetite table (median vs 75th percentile vs largest dip).
- Smallest cuts to stay out of overdraft (a linear program), a backup plan ranked by months gained, emergency-fund milestones, and an AI-written plan in plain English.

05 Why trust it

Headline proof; our runway vs the judges' formula (tested); model health (10 monitors, no breaches); the 3-year backtest; calibration; customers steer back; survival (when first overdrafts come); champion vs challenger; loan rule; fairness; who the riskiest are; which reasons hold up; what didn't hold up and what we did; limits with the production fix.

Pitch mode

Press **Pitch mode** in the top bar. It drives the app through five steps; move with Next/Back or the arrow keys, Esc to leave. Each step shows a short cue of what to point at. The cues contain no numbers, so they can never contradict the screen.

4. The business logic

Who it's for

A bank's **customer-care advisors** (who to call this week, why, and what to offer) and its **risk teams** (how the book behaves, whether a loan should be reviewed). The customer benefits through them: fewer overdrafts, fewer fees, earlier help.

The whole book: every customer gets one tier

Tier	Rule (data up to the as-of month)	Action	Overdrew next year (Dec 2024 → 2025)
Already overdrawn	Any day below $0 in the last 12 months	Recovery plan + personal alert	60%
Act now	Trough in the riskiest 10%	Advisor contact, matched to the cause	12.7%
New and swinging	3–11 months old, top 10% of new accounts by swings in money in	Welcome call + personal alert	18.6%
Watch	Trough in the next 20%	Automatic personal alert, no call	2.7%
Stable	Everyone else	Nothing	0.4% (same every year)
New	Under 3 months, or new but steady	Standard alerts	2.8%

> The headline for a banker: the top three tiers are about **10%** of customers and produce **68%** of next year's overdrafts. More than half the book needs **nothing**.

The action ladder: safest first

1. **Call** when there is a cause to fix (money in fell → income check, EI, payment holiday; a heavy new loan → payment date or longer term).
2. **Alert** at the customer's own danger level. Free, no credit risk. In 2025 it fired before **23 of 34** caught overdrafts, a median **3 months** early (a generic $100 alert: 1 of 34).
3. **Auto-transfer** from their own savings, sized to their typical dip. Free. Covered **27 of 34** caught overdrafts in full; only **$78K** of **$2.9M** was ever drawn: it works as a limit, not cash.
4. **A small overdraft limit**, only after a call, priced at Tangerine's terms ($5 a month used + 19% a year). It comes last because every overdrawn borrower in this data defaulted.

Risk appetite: how big should the buffer be?

Buffer set at	Overdrafts covered, 2025	Limits offered	Actually drawn
Median dip (default)	27 of 34	$2.9M	$78K
75th percentile (conservative)	29 of 34	$3.8M	$83K
90th percentile	29 of 34	$4.8M	$89K
Largest dip	30 of 34	$6.1M	$95K

The 75th percentile is the knee: two more customers covered for about $5K more drawn. The bank chooses; we show the trade-off.

Lending: a check at approval

Hold a loan for review if, using only data before the loan month, the applicant was already overdrawn or is in the riskiest 10% by trough. Nothing is fit on loan outcomes. Result on all 682 loans: holds **118 (17%)**, catches **41 of 76** defaults, held loans default **34.7% vs 6.2%** (counting only loans that finished unpaid: **10.2% vs 3.4%**, p = 0.003). It also holds 77 good loans, so “review” means talk first (smaller loan, cushion, alert), not reject. Dollar figures are amounts lent, not losses.

Why a bank would pay for this

- **Fewer overdrafts and fees** for customers (the federal NSF fee cap is now $10, so fee income is shrinking anyway; preventing overdrafts protects the relationship).
- **Fewer credit losses**: the same signal flags risky loans at approval.
- **Advisor time spent where it matters**: a call list of about 10% of customers, each with a reason.
- **Regulation-ready**: a model card in OSFI E-23 format, monitoring with thresholds, fairness checks, a human in the loop.

The pilot plan (what a consultant asks)

Phase	What happens	Success measure
0–3 months: shadow	Score the whole book monthly; no customer contact; compare flags with real overdrafts	Catch rate and monitors stay within thresholds on Tangerine's own data
3–6 months: controlled pilot	Half of flagged customers get call + alert + transfer; half business as usual	First-overdraft rate, complaints and fees, treated vs control (this proves cause)
6+ months: scale	Call list in the CRM; loan review at origination	Overdrafts and loan losses avoided; advisor time per call

Finance terms: what we added and what we refused

Term	Decision	Why
Haircut	Added	It's the trough in a banker's words: 93172 takes a 99% haircut at the low point
Confidence level / risk appetite	Added, backtested	The buffer table above
Volatility	Already in	Cash-flow swings pick the auto-save action; swings in money in score new accounts
Probability of default (PD)	Have it	Calibrated risk: predicted 1.61% vs 1.61% actual in 2025
Exposure	Have it	Money drawn on the buffer; exposure profile on the Stress screen
Expected loss (PD × EAD × LGD)	Refused	Loss given default isn't in the data; we won't invent it
Portfolio VaR / correlation	Refused	No shared shocks in the data to estimate correlation; the bank-wide job-loss curve shows one shared shock instead

5. The technical guide

Architecture

One rule shapes everything: **all maths lives in engine.py**. Everything else only moves or displays numbers.

Layer	File(s)	Job
Data	data/*.csv (never committed)	9 Tangerine/PKDD'99 files: 4,500 accounts, 185,326 account-months, 1,056,320 transactions, 682 loans
Engine	engine.py	Every calculation: both runways, trough, simulation, calibration, verdict, cushion, diagnosis, loan check, tiers, survival, optimiser
Evidence	evidence.py → evidence.json	Every backtest and statistical test; rebuilt with python evidence.py
Context	ui/context.py, ui/data.py	Builds one customer's numbers (shared by both front ends); caches anything that loops over all accounts
AI	ai.py	Twist-card reader and plan/brief/message writer, with number and wording checks; Gemini on Vertex AI or offline
API	api.py (FastAPI)	JSON endpoints for the dashboard; also serves the built React files
Dashboard	web/ (React 19, Vite, TanStack Query, Recharts)	The five screens and Pitch mode; never calculates money
Backup app	app.py (Streamlit)	Same engine, older UI; used if the React build fails
SQL	sql/trough.sql	The trough as one warehouse query; tested identical to the Python
Ship	Dockerfile, docs/DEPLOY.md, scripts/smoke.py	One container for Cloud Run; smoke test against any URL

How one request flows

1. The browser asks GET /api/customer/93172?asof=2024-12.
2. api.py loads the data once (cached), cuts it at 2024-12 and calls ui/context.build_context.
3. build_context calls engine functions: profile, runway, realistic_profile, liquidity_trough, overdraft_risk, cushion, diagnose, loan_check, when_overdraft…
4. The API returns JSON; the React screen formats and draws it. If the user asks for words, ai.py gets the engine's facts and must reuse them exactly.

Every calculation, defined

Name (engine)	Definition
Judges' runway (profile, runway)	I = median money in, S = median money out, B = mean average daily balance over the last 6 rows, H = median household bills, F = median bills + insurance + loans. Runway = (B − one-off) ÷ (spending after shock − income after shock); 0 if buffer ≤ 0; “no shortfall (120+)” if the gap ≤ 0; cap 120; 1 decimal. Answer key: 20111 → 3.4 / 4.0 / 3.4 months.
Realistic runway (realistic_profile,	Last 12 months' means; pension and interest kept in a job loss; EI 55% capped at $729/week (≈ $3,159/month) and stopping after its months; usable buffer = B

Name (engine)	Definition
realistic_runway)	− typical in-month dip; a loan granted in the window counts at its full payment.
Liquidity trough (ltr_table)	Lowest ÷ average balance over 6 months; scored with 12+ months of history; percentile rank, 0 = most fragile; flag = bottom 10%. Haircut = 1 − trough.
Monte Carlo (simulate)	2,000 futures built from 3-month blocks of the customer's own last 24 months (block bootstrap), the shock applied, starting from today's balance, seeded by account so it's repeatable. Under a shock it reports shares of futures, never “chance”.
Calibrated risk (overdraft_risk)	The raw no-shock 12-month break share mapped through a 2-parameter logistic (Platt scaling) fitted 2023 → 2024, judged on 2025. Raw it was 8× too pessimistic because real customers steer back after a bad month.
Verdict (verdict)	Can they make 6 months: Yes / No / Only on paper (formula yes, realistic no) / Fragile (yes, but many futures fail); the biggest reason is found by switching each adjustment back and measuring the change in months.
Cushion (cushion)	Buffer = alert level = median monthly dip (average − lowest balance); conservative = 75th percentile; top-up now; usual dip day if 70% of months fall within ±5 days.
Diagnosis (diagnose)	In order: overdrawn this year → money in last 3 months ÷ prior 9 ≤ 0.75 → new loan payment ≥ 30% of money in → swings (sd of net flow ÷ balance ≥ 0.47) → dip only. Thresholds fixed on 2023.
Loan check (loan_check)	Hold for review if, before the loan month, ever overdrawn or in the riskiest 10% by trough.
Tiers (book_tiers)	As in Part 4; new accounts scored by the standard deviation of monthly money in, opening month dropped.
Survival (survival_cohort, when_overdraft)	Never-overdrawn customers, clock starts in January, event = first day below $0, up to 24 months, censoring when data stops. The per-customer “When?” uses the cohort scored two years before the as-of month, so it never looks ahead.
Optimiser (optimize_cuts)	One linear program (HiGHS) over 12 months: the smallest monthly cuts that keep the bad-case (10th percentile) path above the dip; bills and insurance can't be cut; solved once (not MPC).
Overdraft cost (overdraft_cost)	$5 per month used + 19% a year (Tangerine's published terms).

Statistics: how we know it's true

Method	Why	Result
Out-of-time backtest	Features to December, outcome the next year, three separate years	Risk ratio 17.9× / 12.8× / 14.6×; AUC 0.84 / 0.84 / 0.86
Bootstrap confidence intervals	How sure each number is	e.g. realistic beats the formula by +0.058 AUC (95% CI 0.031–0.089)
Logistic regression with controls	Is it just balance size?	Odds ratio 16.8 (95% CI 9.5–29.7) after balance, account age, unemployment
Benjamini–Hochberg (q = 0.10)	We tried many ideas; correct for luck	48 of 57 survive; failures shown on screen
Platt calibration + Brier score	Make probabilities honest	1.61% predicted vs 1.61% actual; Brier 0.0142 beats 0.0158
Permutation test (shuffled months)	Do customers steer back?	Yes: +36–37 points vs +18–19 shuffled

Method	Why	Result
Kaplan–Meier + discrete-time hazard	When do overdrafts come?	Flagged: hazard ratio 11.2 and 12.4; 2–4× higher in the first 6 months
Champion vs challenger (gradient boosting)	Would ML do better?	39 vs 34 of 55, not significant (McNemar p = 0.18); its top feature is the trough → keep the trough
Fisher exact test	Small-count comparisons	Loan rule p = 0.003; 27 of 27 loans to overdrawn customers defaulted, p = 1.6e-28
Four-fifths rule + Fisher	Fairness by gender and age	Flag-rate ratio 1.12 and 0.86; catch-rate gap flips direction, not significant
PSI drift + thresholds	Ongoing monitoring	10 monitors, 0 breaches

The AI, and how it is kept honest

- **Model:** Gemini 3.5 Flash on Vertex AI (the event's Google Cloud project). No key is stored: the machine's or service's Google identity is used. Without it, everything still works offline (rules + templates).
- **Reading twist cards:** Gemini returns typed JSON only; values are clamped; the offline rules read the same card and any disagreement is flagged on the chips before anything runs.
- **Writing:** the brief, the customer message (English/French, no risk percentages, no credit offer) and the plan. Every number must appear in the engine's facts (verify_numbers) and simulation shares can't be called a “chance” (verify_wording); otherwise a template is used.
- **Speed limit:** 6 seconds, then rules or template (one call once hung for 92 seconds; fixed).
- **Measured:** on 16 held-out cards never used for tuning, Gemini 14/16 (88%), offline rules 15/16 (94%). Neither is perfect alone, which is why the chips come first.

The API

Endpoint	Returns
GET /api/health	OK, which AI is active, 4,500 accounts
GET /api/customer/{acct}?asof	Everything on Know the customer
POST /api/shock/{acct}	Both runways, why they differ, verdict, simulation
POST /api/plan/{acct}	Cushion, costs, optimiser, backup levers, milestones, facts for the AI
GET /api/watchlist?asof, /api/book?asof	The call list and the whole-book tiers
POST /api/twist	Settings, chips and disagreements from a twist card
POST /api/ai/brief, /api/ai/message, /api/write-plan	AI text (checked) or a template
POST /api/portfolio, GET /api/evidence	Bank-wide job-loss curve; every tested result

Tests and the software lifecycle

Stage	Check	Count / result
Engine	pytest -q: acceptance tests per calculation, no-lookahead guards, answer key	123 fast tests
Statistics	pytest -m slow: every evidence claim re-proven	14

Stage	Check	Count / result
API ↔ screens	Contract tests; a test parses the React screens for every evidence section they read	21 API tests
AI	Fake-model tests + scripts/eval_ai.py on tuned, dev and held-out cards	17 + eval report
Screens	Streamlit click-through; React type-check and build; axe-core accessibility	6; 0 WCAG violations
CI	GitHub Actions on every pull request: compile, data-free tests, dashboard build	Green on every PR
Release	Docker image for linux/amd64 + scripts/smoke.py (8 checks) against it	See Part 8
Operate	Monitoring with thresholds on the Why trust it screen; model card (docs/MODEL_CARD.md)	0 breaches

The full list of test cases for each financial calculation is in docs/TEST_PLAN.md.

Docker and the cloud

One container: a Node stage builds the React app, then a Python stage installs the engine and serves both the API and the built pages with uvicorn on port 8080. The data is copied into the image at build time (never into git). Commands are in docs/DEPLOY.md:

- **Local:** docker build --platform linux/amd64 -t runway:local . then docker run -p 8080:8080 runway:local, then python scripts/smoke.py http://localhost:8080.
- **Cloud Run:** gcloud run deploy runway --source . with 4 GB memory, 2 CPUs, one warm instance, at most three; Vertex permission given to the service's identity.
- **What actually happened:** --source failed on build-account permissions, so the image was built on the laptop, pushed to the project's Artifact Registry and deployed with --image (no extra permissions). It runs on the event's Google Cloud credit; one warm copy (2 CPU, 4 GB, at most 3) costs credit every hour, so watch the billing page.
- **After the event:** delete the service and the project.

6. What you built

You (Sukhman) led the engine: the maths, the statistics, the evidence, and the repository. The work in the order it happened:

Round	What was built	Why it mattered
Foundation	Judges' formula matching the README exactly; realistic runway (EI cap and end, pensions, dip)	The answer key is right, and our runway beats it out of time
The finding	Liquidity trough; 3-year out-of-time backtest; confounders; multiple-testing correction	A discovery, not a dashboard
Forecast	Monte Carlo, found 8× too pessimistic, explained (customers steer back), calibrated	Honest probabilities, and a second finding
Action	Diagnosis → next best action; cushion and personal alert backtested	Tells the advisor what to do
Credit	Loan-approval rule; forward loan payments; Time Machine leak fixed	Business impact in dollars lent; the 93172 story
Governance	Monitoring, fairness, model card (OSFI E-23), CI, SQL proof	What a bank needs before production
Product	React dashboard, Pitch mode, story strip, call list, whole-book tiers, accessibility	Judges understand it in 5 seconds
AI	Gemini on Vertex with typed output, checks, timeout, fallback, measured evaluation	AI that can't invent numbers
Depth	Survival analysis, champion vs challenger, new-account score, buffer risk appetite, haircut	Answers “when?”, “why not ML?”, “what about new customers?”
Ship	Docker image, Cloud Run guide, smoke test, test plan, this guide	A product, not a notebook

7. Showing it to the judges

Before you walk up

- Laptop on power; server running; browser at the app, Time Machine on Dec 2024, account 93172.
- Top bar says “AI: Gemini on Vertex AI”. If the Wi-Fi is bad it falls back to offline rules: still read the chips before running.
- Know three accounts by heart: **93172** (the story), **20111** (the answer key: 3.4 / 4.0 / 3.4 months), **20176** (a second twist).

The 60-second pitch (Pitch mode, press → five times)

1. **The finding:** banks watch the average; overdrafts happen at the low point. Flag 10% → catch 34 of 55 first overdrafts, three years running.
2. **One customer (93172, Dec 2024):** looks fine ($18,503 average, “no shortfall”) → dipped to $238 → the bank lent $167K that month → realistic runway 2.4 months → overdrawn in January. A trough check at approval would have held the loan.
3. **Twist card:** read the verdict, then “what moves the answer”.
4. **Plan:** fix the cause first (call), then the no-risk steps (alert, transfer from savings); a limit comes last, with its real cost.
5. **Trust:** tested on years the model never saw; model health shows no breaches; what failed is listed.

The live twist (3 minutes, judge picks the account)

1. Type their account in the box → Open.
2. Go to 03 Stress test → paste or type their twist card.
3. Point at the chips: “this is what the AI read; the rules agree (or: here's the disagreement)”. Confirm.
4. Read the verdict sentence out loud, then the biggest reason in months.
5. Scroll to the two runways: “the judges' formula says X, realistically Y, because…”.
6. Go to 04 Make a plan: “first we'd do this; it costs them $0”.

What to touch, and what not to

Touch	Avoid unless asked
Pitch mode; the story strip; the verdict; the chips; “Do this first”; the whole-book card; the loan card	The optimiser chart (big cuts look unrealistic); the exposure profile (uncalibrated); the full trait table; Streamlit

Hard questions, one-line answers

Question	Answer
Isn't the trough just a low balance?	Yes, that's the point: the standard runway and the average both miss it. The ratio and the plain low point are tied; both beat both runway formulas.
Why offer credit to risky customers?	We don't lead with it: alert and call first, then a transfer from their own savings ($0); a limit only after a call, at Tangerine's terms.

Question	Answer
False positives?	The alert pinged 22% of flagged customers who never went overdrawn; 1 in 4 alerts was followed by a real overdraft vs a 1.6% base rate.
Is it fair?	Flag rates within the four-fifths band both years; the catch-rate gap flips direction and isn't significant. Gender and age are never inputs.
Why not ML?	We tested it: gradient boosting caught 39 vs 34, not significant, and its top feature was the trough. One explainable ratio stays champion.
When should the bank act?	In the first 6 months: flagged customers have 11–12× the hazard, highest early.
Can the AI make up a number?	No: every number must be in the engine's facts or a template is used. Measured accuracy on unseen cards: 14 of 16.
Does it scale?	The core signal is one SQL window query, tested identical to the Python on all 4,500 accounts.
Is it causal?	No, it's a signal; the pilot's controlled test is how you'd prove the actions work.
Biggest weakness?	Money in is mostly cash, so a job loss assumes all non-pension money stops; and it's 1990s data, which is why the pilot starts in shadow mode.

8. Is it ready to ship?

Ready for the datathon: yes

Check	Status
All work merged to main; CI green	Yes (PRs #13–#17)
Tests: fast, UI, slow, CI mode	All pass
Evidence rebuild reproduces evidence.json	Yes (rounding only)
Accessibility (WCAG 2 A/AA)	0 violations on all five screens
Phone width	No horizontal scroll at 375 px
Offline fallback	Works without Wi-Fi (rules + templates)
Docker image and smoke test	Image builds for linux/amd64; scripts/smoke.py passes 8 of 8 against it
Live link (Cloud Run, event credits)	https://runway-1085585470420.us-central1.run.app: smoke test 8 of 8, Gemini on Vertex
Full QA run (docs/QA_REPORT.md)	201 customer-dates, 24 page checks, Pitch mode: no broken values, no n/a, 0 accessibility violations

Ready for a real bank: not yet (say this before a CTO does)

- No sign-in, roles or audit log; anyone with the link sees every account.
- No customer-data protections (encryption, masking, retention rules).
- One container, data baked in; a real deployment reads from the warehouse (the SQL version is the start).
- Trained on 1990s Czech data: it must be re-validated on Tangerine's own data in shadow mode first.
- No causal proof that the actions reduce overdrafts until the controlled pilot.

9. Strict rating and ranking

Every component: is it useful?

Component	Score /10	Verdict
Liquidity trough early warning	9.5	The core finding: simple, tested three years out of time, survives controls
Realistic runway (EI cap/end, pensions, dip, new loans)	9	Beats the judges' formula out of time; powers the 93172 story
Loan-approval rule	9	Clear dollar-framed impact for bankers
Pitch mode + story strip	9	Turns a dense product into a 60-second story
Whole-book tiers + call list	8.5	Makes it a bank tool, not a single-customer toy
Verdict + biggest reason	8.5	Best answer to the live twist
Cushion, alert, risk-appetite table	8.5	Backtested actions with real costs
Calibrated Monte Carlo	8	Only honest way to say “13.8%”; raw it was wrong, which became a finding
Governance: monitoring, fairness, model card, CI	8.5	Rare in a datathon; exactly what banks ask
AI with number checks + evaluation	8	Safe and measured; deliberately small
Champion vs challenger	8	Answers “why not ML?” with numbers
Survival analysis (“When?”)	7.5	Rigorous, nice for the professor; doesn't change a decision
Diagnosis reasons	7	Hold in 2025, not 2024; used to pick actions, not claimed as proven
New-account score	7	Closes a real blind spot; smaller samples
SQL scale proof	7.5	Short, convincing answer for data-platform judges
Overdraft cost at Tangerine terms	7.5	Small, but makes “credit last” concrete
Linear-program cuts	6	Mathematically right; the cut sizes can look unrealistic (93172 at Dec 2024: about $7,330 a month on average, up to $11,690); folded behind a click
Exposure profile	5	Uncalibrated, niche; keep it folded away
Streamlit backup app	4	Insurance only; never demo it
Docker + Cloud Run	7	Works as one container; untested at load

The judges' view

Criterion (weight)	Score	Why
Discovery (35%)	8.5	Several tested findings: the low point, customers steer back, hidden new loans, the loan rule; failures disclosed
Data use (25%)	9.5	All 9 files used; leaks caught and tested; out-of-time on three years; fairness; SQL identical to Python
Communication (25%)	8	Pitch mode and the story strip make it legible; the rest depends on rehearsal

Criterion (weight)	Score	Why
Impact (15%)	8.5	Call list, loan rule, real pricing, risk appetite, pilot plan; no “dollars saved” figure (we don't have loss data)
Weighted	≈ 8.7 / 10	

Where we stand

> Honest ranking: on substance this is very likely **top 3 in the Tangerine track, with a real shot at 1st**. I can't see the other entries, so this is a judgement, not a measurement. Few datathon teams ship out-of-time validation, calibration, multiple-testing correction, fairness checks and a model card.

What can still lose it isn't the code: a fumbled live twist, a demo laptop without data or sign-in, a rushed video, or too many numbers in the pitch. The best use of the remaining time is **rehearsal**: Pitch mode five times, and the live twist on 93172, 20111 and 20176 until it's boring.